Radiological Hazard Assessment Tool

Technical Documentation and User Guide

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Abstract

This document provides comprehensive technical documentation for the Radiological Hazard Assessment Tool, a Python-based desktop application developed for computing radiological hazard indices from environmental radiation measurements. The application implements standard radiological equations with full uncertainty propagation, providing researchers and environmental scientists with a robust tool for assessing radiation hazards from naturally occurring radioactive materials (NORM). The tool features batch processing capabilities, interactive data visualization with uncertainty error bars, and comprehensive results export functionality.

Contents

1 Introduction

1.1 Overview

The Radiological Hazard Assessment Tool is a desktop application designed to compute a suite of standard radiological hazard indices from measurements of naturally occurring radioactive materials (NORM). The application processes activity concentrations of three key radionuclides:

- **Radium-226 (Ra-226)**: A decay product of uranium series
- **Thorium-232 (Th-232)**: A decay product of thorium series
- **Potassium-40 (K-40)**: A naturally occurring radioactive isotope of potassium

1.2 Application Purpose

The tool serves as a comprehensive solution for:

- Computing five standard radiological hazard indices
- Propagating measurement uncertainties through all calculations
- Classifying samples as safe or unsafe based on international thresholds
- Visualizing relationships between radionuclide concentrations and dose rates
- Batch processing large datasets
- Exporting results for further analysis

1.3 Target Users

- Environmental scientists
- Radiation protection professionals
- Researchers in radiochemistry and health physics
- Students in nuclear and radiological sciences

1.4 Key Features

1. **Batch Processing**: Load and process multiple samples simultaneously
2. **Uncertainty Propagation**: Full uncertainty analysis following standard error propagation
3. **Interactive Visualization**: Correlation plots with error bars and regression analysis
4. **Safety Classification**: Automatic classification based on international thresholds
5. **Data Export**: CSV export of results for external analysis
6. **Plot Export**: High-quality plots in PNG, PDF, or SVG formats

2 Mathematical Framework

2.1 Computed Hazard Indices

The application computes five standard radiological hazard indices. Table 1 summarizes these indices, their symbols, formulas, and safety thresholds.

Table 1: Radiological Hazard Indices and Safety Thresholds

Index	Symbol	Formula	Threshold
Radium Equivalent	Rad_Eq	$Ra_{eq} = C_{Ra} + 1.43C_{Th} + 0.077C_K$	≤ 370 Bq/kg
External Hazard	Hex	$H_{ex} = \frac{C_{Ra}}{370} + \frac{C_{Th}}{259} + \frac{C_K}{4810}$	≤ 1
Internal Hazard	Hin	$H_{in} = \frac{C_{Ra}}{185} + \frac{C_{Th}}{259} + \frac{C_K}{4810}$	≤ 1
Dose Rate	Dose_Rate	$\dot{D} = 0.462C_{Ra} + 0.604C_{Th} + 0.0417C_K$	≤ 59 nGy/h
AEDE	AEDE	$AEDE = \dot{D} \times 8760 \times 0.2 \times 10^{-6}$	≤ 1 mSv/y

2.2 Detailed Mathematical Formulation

2.2.1 Radium Equivalent Activity

The radium equivalent activity (Ra_{eq}) is a weighted sum of the three radionuclides, normalized to radium-226:

$$Ra_{eq} = C_{Ra} + 1.43C_{Th} + 0.077C_K \tag{1}$$

where C_{Ra}, C_{Th}, and C_K are the activity concentrations in Bq/kg.

2.2.2 External Hazard Index

The external hazard index (H_{ex}) estimates the external radiation hazard:

$$H_{ex} = \frac{C_{Ra}}{370} + \frac{C_{Th}}{259} + \frac{C_K}{4810} \tag{2}$$

2.2.3 Internal Hazard Index

The internal hazard index (H_{in}) accounts for internal radiation exposure:

$$H_{in} = \frac{C_{Ra}}{185} + \frac{C_{Th}}{259} + \frac{C_K}{4810} \tag{3}$$

2.2.4 Absorbed Dose Rate

The absorbed dose rate ($\dot{D}$) in air at 1 meter above ground:

$$\dot{D} = 0.462C_{Ra} + 0.604C_{Th} + 0.0417C_K \quad \text{[nGy/h]} \tag{4}$$

2.2.5 Annual Effective Dose Equivalent

The annual effective dose equivalent (AEDE) for outdoor exposure:

$$AEDE = \dot{D} \times 8760 \times 0.2 \times 10^{-6} \quad \text{[mSv/y]} \tag{5}$$

where:

- 8760 = number of hours in a year
- 0.2 = outdoor occupancy factor
- 10^{-6} = conversion factor from nGy to mGy

2.3 Uncertainty Propagation

The application implements rigorous uncertainty propagation following the law of propagation of uncertainty for independent variables:

$$\sigma_f = \sqrt{\sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}\right)^2 \sigma_{x_i}^2} \tag{6}$$

2.3.1 Radium Equivalent Uncertainty

$$\sigma_{Ra_{eq}} = \sqrt{(1 \cdot \sigma_{Ra})^2 + (1.43 \cdot \sigma_{Th})^2 + (0.077 \cdot \sigma_K)^2} \tag{7}$$

2.3.2 External Hazard Uncertainty

$$\sigma_{H_{ex}} = \sqrt{\left(\frac{\sigma_{Ra}}{370}\right)^2 + \left(\frac{\sigma_{Th}}{259}\right)^2 + \left(\frac{\sigma_K}{4810}\right)^2} \tag{8}$$

2.3.3 Internal Hazard Uncertainty

$$\sigma_{H_{in}} = \sqrt{\left(\frac{\sigma_{Ra}}{185}\right)^2 + \left(\frac{\sigma_{Th}}{259}\right)^2 + \left(\frac{\sigma_K}{4810}\right)^2} \tag{9}$$

2.3.4 Dose Rate Uncertainty

$$\sigma_{\dot{D}} = \sqrt{(0.462 \cdot \sigma_{Ra})^2 + (0.604 \cdot \sigma_{Th})^2 + (0.0417 \cdot \sigma_K)^2} \tag{10}$$

2.3.5 AEDE Uncertainty

$$\sigma_{AEDE} = 8760 \times 0.2 \times 10^{-6} \times \sigma_{\dot{D}} \tag{11}$$

2.4 Safety Classification

Samples are classified as "Safe" or "Unsafe" based on the following criteria:

$$\text{Safe} = \begin{cases} \text{True} & \text{if } Ra_{eq} \leq 370 \text{ and } H_{ex} \leq 1 \text{ and } H_{in} \leq 1 \text{ and } \dot{D} \leq 59 \text{ and } AEDE \leq 1 \\ \text{False} & \text{otherwise} \end{cases} \tag{12}$$

The application also calculates a safety probability using the uncertainty distribution:

$$P_{\text{safe}} = \Phi\left(\frac{\text{Threshold} - \text{Value}}{\text{Uncertainty}}\right) \tag{13}$$

where Φ is the standard normal cumulative distribution function.

3 Software Architecture

3.1 Technology Stack

Table 2: Technology Stack

Component	Technology	Purpose
Language	Python 3	Application development
GUI Framework	PyQt5	User interface
Data Processing	pandas	CSV handling and data manipulation
Numerical Computing	NumPy	Mathematical operations
Visualization	Matplotlib	Plotting and graphics

3.2 System Architecture

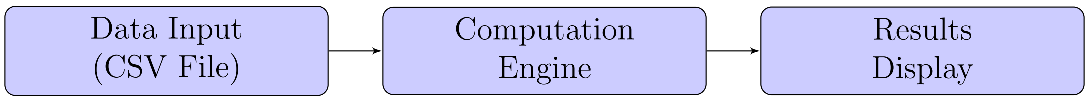

Figure 1: High-level System Architecture

3.3 Core Components

3.3.1 1. Computation Engine

The computation engine is a pure function that:

- Takes a pandas DataFrame with radionuclide concentrations and uncertainties
- Computes all five hazard indices
- Propagates uncertainties using equations 7 to 11
- Returns a DataFrame with values and uncertainties in separate columns

3.3.2 2. Graphical User Interface

The GUI consists of two main panels:

- **Left Panel**: Controls and input options
- **Right Panel**: Tabbed display for plots and results

3.3.3 3. Plotting Module

Generates correlation plots with:

- Error bars showing 1σ uncertainty
- Linear regression lines with R^2 values
- Proper axis scaling and legends

3.4 Data Flow

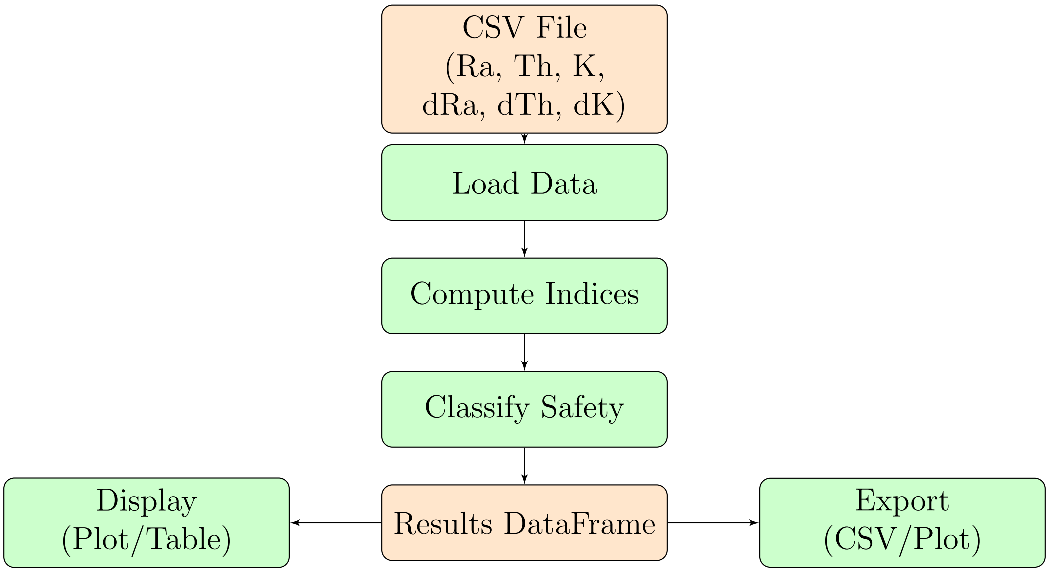

Figure 2: Data Flow Diagram

4 User Interface Guide

4.1 Main Window Layout

```
+-----------------------------------------------------------------+
|  Radiological Hazard Assessment Tool                               |
+------------------------------------------+----------------+
|  Batch Processing                           |   TAB VIEW      |
|  +-----------------------------------+      |               |
|  | Data Input                        |      | [Plot] [Table]|
|  | [Load CSV] No file loaded         |      |               |
|  | Samples: 0                        |      |               |
|  +-----------------------------------+      |               |
|  | CSV Format                        |      |               |
|  | Required: Ra, Th, K, dRa, dTh, dK|     |               |
|  +-----------------------------------+      |               |
|  | Actions                           |      |               |
|  | [Process Samples]                 |      |               |
|  | Progress: [====>]                 |      |               |
|  | [Export CSV] [Save Plot]          |      |               |
|  +-----------------------------------+      |               |
|  | Plot Options                      |      |               |
|  | [Generate Plot]                   |      |               |
|  +-----------------------------------+      |               |
|  | Safety Thresholds                 |      |               |
|  | Rad_Eq <= 370                     |      |               |
|  | Hex <= 1                          |      |               |
|  +-----------------------------------+      |               |
|  Status: Ready                              |               |
+------------------------------------------+----------------+
```

4.2 Controls Panel Description

4.2.1 Data Input Section

- **Load CSV**: Opens file dialog to select a CSV file
- **File Label**: Displays the loaded filename
- **Samples Info**: Shows number of samples loaded

4.2.2 Actions Section

- **Process Samples**: Initiates computation for all samples
- **Progress Bar**: Shows processing progress
- **Export CSV**: Saves results to CSV file
- **Save Plot**: Saves current plot as image

4.2.3 Plot Options Section

- **Generate Plot**: Creates/updates the correlation plot

4.2.4 Safety Thresholds Section

Displays the international safety thresholds for all five indices.

4.3 Results Display

4.3.1 Plot Tab

Shows the correlation plot with:

- Scatter plot of Dose Rate vs each radionuclide
- Error bars showing 1σ uncertainty
- Linear regression lines
- R^2 values for each correlation
- Proper axis labels and legend

4.3.2 Table Tab

Displays results in a tabular format with:

- Sample ID
- All computed indices with uncertainties
- Safety status (color-coded)
- Alternating row colors for readability

5 Technical Implementation

5.1 Core Functions

5.1.1 compute_indices Function

This is the primary computational function that implements equations 1 through 11.

```python
def compute_indices(df):
    """
    Compute hazard indices with uncertainty propagation.
    Returns DataFrame with value and uncertainty in separate columns.
    """
    df = df.copy()

    # Extract uncertainties (default 0 if not present)
    dRa = df.get("dRa", pd.Series([0] * len(df)))
    dTh = df.get("dTh", pd.Series([0] * len(df)))
    dK = df.get("dK", pd.Series([0] * len(df)))

    # Calculate values
    rad_eq = df["Ra"] + 1.43 * df["Th"] + 0.077 * df["K"]
    hex_val = df["Ra"]/370 + df["Th"]/259 + df["K"]/4810
    hin_val = df["Ra"]/185 + df["Th"]/259 + df["K"]/4810
    dose_rate = 0.462*df["Ra"] + 0.604*df["Th"] + 0.0417*df["K"]
    aede = dose_rate * 8760 * 0.2 * 1e-6

    # Propagate uncertainties
    dRadEq = np.sqrt((1*dRa)**2 + (1.43*dTh)**2 + (0.077*dK)**2)
    dHex = np.sqrt((dRa/370)**2 + (dTh/259)**2 + (dK/4810)**2)
    dHin = np.sqrt((dRa/185)**2 + (dTh/259)**2 + (dK/4810)**2)
    dDose = np.sqrt((0.462*dRa)**2 + (0.604*dTh)**2 + (0.0417*dK)**2)
    dAEDE = dDose * 8760 * 0.2 * 1e-6

    return pd.DataFrame({
        "Rad_Eq": rad_eq, "dRad_Eq": dRadEq,
        "Hex": hex_val, "dHex": dHex,
        "Hin": hin_val, "dHin": dHin,
        "Dose_Rate": dose_rate, "dDose_Rate": dDose,
        "AEDE": aede, "dAEDE": dAEDE
    })
```

Listing 1: Computation Engine Implementation

5.1.2 CSV Loading Function

Handles file loading with automatic zero-fill for missing uncertainty columns.

```python
def load_csv(self):
    path, _ = QFileDialog.getOpenFileName(
        self, "Select CSV File", "", "CSV Files (*.csv)"
    )
    if not path:
        return

    try:
        df = pd.read_csv(path)
        required = ["Ra", "Th", "K", "dRa", "dTh", "dK"]
```

```

        # Add zeros for missing uncertainty columns
        missing = []
        for col in required:
            if col not in df.columns:
                df[col] = 0.0
                missing.append(col)

        if missing:
            self.status.setText(f"Columns set to zero: {', '.join(missing)}")

        self.data = df[required]
        # Update UI...
    except Exception as e:
        # Handle error...
```

Listing 2: CSV Loading Function

5.2 Plotting Implementation

The correlation plot is generated using Matplotlib with error bars:

```
def plot_correlation(self):
    self.figure.clear()
    ax = self.figure.add_subplot(111)

    # Extract data
    Ra = self.data["Ra"].values
    Dose = self.results["Dose_Rate"].values
    dDose = self.results["dDose_Rate"].values

    # Sort for clean plotting
    idx = np.argsort(Ra)
    Ra_s = Ra[idx]
    Dose_s = Dose[idx]
    dDose_s = dDose[idx]

    # Error bar plot
    ax.errorbar(Ra_s, Dose_s, yerr=dDose_s, fmt='o',
                color='blue', capsize=3, label='Ra')

    # Regression line
    fit = np.polyfit(Ra, Dose, 1)
    line = np.poly1d(fit)
    ax.plot(Ra_s, line(Ra_s), '--', color='blue')

    # R-squared value
    r2 = np.corrcoef(Ra, Dose)[0,1]**2
    ax.text(0.05, 0.95, f'R$^2$ = {r2:.3f}', transform=ax.transAxes)

    # Labels and grid
    ax.set_xlabel('Concentration (Bq/kg)')
    ax.set_ylabel('Dose Rate (nGy/h)')
    ax.grid(True)

    self.canvas.draw()
```

Listing 3: Plot Generation Function

5.3 Safety Classification

```
indices["Status"] = indices.apply(
    lambda row: "Safe" if all([
        row["Rad_Eq"] <= 370,
        row["Hex"] <= 1,
        row["Hin"] <= 1,
        row["Dose_Rate"] <= 59,
        row["AEDE"] <= 1
    ]) else "Unsafe",
    axis=1
)
```

Listing 4: Safety Classification

6 Installation and Setup

6.1 System Requirements

Table 3: System Requirements

Component	Requirement
Operating System	Windows 10+, Linux, macOS
Python Version	Python 3.7 or higher
RAM	4 GB minimum, 8 GB recommended
Disk Space	100 MB for application and dependencies

6.2 Required Python Packages

```
pip install PyQt5 pandas numpy matplotlib scipy
```

Listing 5: Required Packages Installation

6.3 Installation Steps

1. Ensure Python 3.7+ is installed on your system
2. Install required packages using pip
3. Download the application source code
4. Run the application:

   ```
   python main.py
   ```

6.4 Creating a Standalone Executable

```
pip install pyinstaller
pyinstaller --onefile --windowed main.py
```

Listing 6: Creating Executable with PyInstaller

7 CSV File Format

7.1 Required Format

The application expects CSV files with the following columns:

Table 4: CSV Column Specifications

Column	Unit	Description
Ra	Bq/kg	Activity concentration of Ra-226
Th	Bq/kg	Activity concentration of Th-232
K	Bq/kg	Activity concentration of K-40
dRa	Bq/kg	Standard uncertainty for Ra-226
dTh	Bq/kg	Standard uncertainty for Th-232
dK	Bq/kg	Standard uncertainty for K-40

7.2 Example CSV File

```
Ra,Th,K,dRa,dTh,dK
25.3,12.1,150.0,1.2,0.8,12.0
18.7,9.8,120.5,0.9,0.6,10.5
32.1,15.4,180.2,1.5,1.0,14.4
```

Listing 7: Example CSV Format

7.3 Notes on CSV Format

- Headers are case-sensitive
- Missing uncertainty columns are automatically set to zero
- All numeric values should be in decimal format
- Comments or metadata rows are not supported

8 Output and Results

8.1 Results Table Format

The results table includes the following columns:

Table 5: Results Table Columns

Column	Value	Uncertainty	Unit	Threshold
Sample	-	-	-	-
Rad_Eq	X	σ_X	Bq/kg	≤ 370
Hex	X	σ_X	-	≤ 1
Hin	X	σ_X	-	≤ 1
Dose_Rate	X	σ_X	nGy/h	≤ 59
AEDE	X	σ_X	mSv/y	≤ 1
Status	Safe/Unsafe	-	-	-

8.2 CSV Export Format

The exported CSV file includes:

- All computed indices with their uncertainties
- Sample identification numbers
- Safety status classification
- Proper header labels

8.3 Plot Output

The correlation plot includes:

- Scatter plots with error bars (1σ)
- Linear regression lines
- R-squared values
- Axis labels with units
- Professional dark theme

9 Troubleshooting and FAQ

9.1 Common Issues

9.1.1 File Loading Issues

- **Issue**: CSV file not loading
- **Solution**: Check that column headers match exactly (Ra, Th, K, dRa, dTh, dK)
- **Solution**: Ensure all values are numeric

9.1.2 Processing Issues

- **Issue**: Computation error during processing
- **Solution**: Check for non-numeric values in input data
- **Solution**: Verify that uncertainty columns contain valid numbers

9.1.3 Plotting Issues

- **Issue**: Plot not displaying
- **Solution**: Click "Generate Plot" after processing
- **Solution**: Ensure data has been processed first

9.2 Frequently Asked Questions

1. **What if my CSV doesn't have uncertainty columns?**
   - The application will automatically add them with zeros
   - A warning message will appear indicating this
2. **How are uncertainties propagated?**
   - Using the standard error propagation formula (Equation 6)
   - All uncertainties are independent and normally distributed
3. **What do the R^2 values on the plot mean?**
   - They indicate the correlation strength between each radionuclide and dose rate
   - Values closer to 1 indicate stronger correlation
4. **How is safety status determined?**
   - A sample is "Safe" only if ALL indices are below their thresholds
   - The safety probability considers the uncertainty distribution

10 Performance and Limitations

10.1 Performance Characteristics

- **Processing Speed**: Can handle thousands of samples in seconds
- **Memory Usage**: Efficient for datasets up to millions of samples
- **Plot Generation**: Near-instantaneous for reasonable sample sizes

10.2 Limitations

- CSV format is fixed and not customizable
- Only supports the five standard indices
- No database connectivity
- No network functionality

10.3 Future Enhancements

Planned features for future versions:

- Additional radiological indices
- Support for Excel files
- Advanced statistical analysis
- Report generation in PDF
- Database integration

11 Code Structure and Documentation

11.1 File Structure

The application is contained in a single Python file with the following structure:

```
main.py
|-- Imports
|-- compute_indices()         # Computation Engine
|-- MainApp Class
|    |-- __init__()           # GUI Setup
|    |-- load_csv()           # Data Loading
|    |-- compute_all()        # Batch Processing
|    |-- plot_correlation()   # Plot Generation
|    |-- save_results()       # CSV Export
|    |-- save_plot()          # Plot Export
|    |-- apply_style()        # CSS Styling
|-- __main__                  # Application Entry Point
```

Listing 8: Code Structure

11.2 Code Documentation Standards

All functions include:

- Docstrings describing purpose
- Parameter descriptions
- Return value specifications
- Exception handling notes

11.3 Error Handling

The application implements comprehensive error handling:

- Try/except blocks for file operations
- Input validation for numerical entries
- Graceful error messages to user
- Exception logging for debugging

12 Appendix

12.1 Mathematical Derivations

12.1.1 Derivation of Radium Equivalent Uncertainty

Starting from equation 1:

$$Ra_{eq} = C_{Ra} + 1.43C_{Th} + 0.077C_K$$

Applying equation 6:

$$\sigma_{Ra_{eq}} = \sqrt{\left(\frac{\partial Ra_{eq}}{\partial C_{Ra}}\right)^2 \sigma_{Ra}^2 + \left(\frac{\partial Ra_{eq}}{\partial C_{Th}}\right)^2 \sigma_{Th}^2 + \left(\frac{\partial Ra_{eq}}{\partial C_K}\right)^2 \sigma_K^2}$$

$$\sigma_{Ra_{eq}} = \sqrt{(1)^2\sigma_{Ra}^2 + (1.43)^2\sigma_{Th}^2 + (0.077)^2\sigma_K^2}$$

$$\sigma_{Ra_{eq}} = \sqrt{\sigma_{Ra}^2 + (1.43\sigma_{Th})^2 + (0.077\sigma_K)^2}$$

12.1.2 Derivation of Dose Rate Uncertainty

Starting from equation 4:

$$\dot{D} = 0.462C_{Ra} + 0.604C_{Th} + 0.0417C_K$$

Applying equation 6:

$$\sigma_{\dot{D}} = \sqrt{(0.462)^2\sigma_{Ra}^2 + (0.604)^2\sigma_{Th}^2 + (0.0417)^2\sigma_K^2}$$

$$\sigma_{\dot{D}} = \sqrt{(0.462\sigma_{Ra})^2 + (0.604\sigma_{Th})^2 + (0.0417\sigma_K)^2}$$

12.2 Reference Standards

Table 6: Reference Standards and Guidelines

Organization	**Reference**
UNSCEAR	United Nations Scientific Committee on the Effects of Atomic Radiation
IAEA	International Atomic Energy Agency Safety Standards
ICRP	International Commission on Radiological Protection
WHO	World Health Organization Guidelines

12.3 Glossary of Terms

Table 7: Glossary

Term	Definition
NORM	Naturally Occurring Radioactive Materials
AEDE	Annual Effective Dose Equivalent
Bq/kg	Becquerel per kilogram (unit of specific activity)
nGy/h	Nanogray per hour (unit of absorbed dose rate)
mSv/y	Millisievert per year (unit of annual effective dose)
Uncertainty	Quantitative measure of measurement precision
Propagation	Process of combining uncertainties through calculations
CSV	Comma Separated Values (file format)
GUI	Graphical User Interface
API	Application Programming Interface

12.4 References

1. UNSCEAR. (2000). Sources and Effects of Ionizing Radiation. United Nations.
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